Teaching Machines to Hear Better Than People

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The crack of a bat and the cheers of a crowd can tell a baseball fan that something exciting happened, even without seeing it. At MIT, Professor Paris Smaragdis wants machines that hear and understand sounds even better than people.

It could hear a traffic accident in squealing tires and crunching metal, or tell when a drill digging a **geothermal** well might break. He says his work does for hearing what **computer vision** does for seeing.

Smaragdis joined MIT in 2025 and leads a group studying **machine learning**, **sound source separation**, and music tools.

Music started his journey. He grew up in Greece, played classical guitar, and later used computers and **synthesizers** at Berklee College of Music in Boston, where he majored in bass.

With mentor Richard Boulanger, he made music tools. Then he worked with Professor Barry Vercoe at the MIT Media Lab.

He then turned to teaching computers to understand sound, choosing **machine learning**. He believes we must teach computers how to listen, because we cannot simply tell them how.

Word Treasure

geothermal -- Geothermal means heat that comes from deep inside the Earth and can be used for energy.

synthesizers -- Synthesizers are electronic instruments that can make many different sounds.

machine learning -- Machine learning is a way for computers to learn from examples instead of being told every single step.

computer vision -- Computer vision is how computers learn to understand pictures and video.

sound source separation -- Sound source separation means picking out one sound from a mix of many sounds happening at once.

Background you need

Machine learning is a way of teaching computers by showing them lots of examples, so they can learn to spot patterns on their own instead of following simple step-by-step orders.

Computer vision does the same kind of job for seeing, helping computers make sense of pictures and video.

Paris Smaragdis wants to bring that idea to hearing, so machines can pick out important sounds, like a crash or a breaking drill, and understand what they mean.

Break it down

WHO: Paris Smaragdis, who leads a group at MIT studying machine learning, sound, and music tools.

WHAT: He is working on machines that hear and understand sounds even better than people.

WHERE: At MIT, after earlier work at Berklee College of Music in Boston and the MIT Media Lab.

WHY: So machines can notice important sounds, like a traffic accident or a drill about to break, and understand what they mean.

Why it matters

If machines learn to listen well, they could one day warn people about dangers like crashes before anyone even sees them.

Test what you remember

1. What does Paris Smaragdis want machines to do even better than people?

- (A) Play classical guitar
- (B) Cheer at baseball games
- (C) Dig geothermal wells
- (D) Hear and understand sounds

2. What is one thing the machines could hear, according to the article?

- (A) A phone ringing at school
- (B) A drill that might break
- (C) A guitar being tuned
- (D) A crowd singing a song

3. What does Smaragdis say his work does for hearing?

- (A) What a radio does for music
- (B) What a hearing aid does for ears
- (C) What computer vision does for seeing
- (D) What a microphone does for singing

4. What did Smaragdis study at Berklee College of Music in Boston?

- (A) Singing
- (B) Drums
- (C) Bass
- (D) Classical guitar

5. Who was Smaragdis's mentor when he made music tools?

- (A) A baseball fan
- (B) Richard Boulanger
- (C) A geothermal engineer
- (D) Barry Vercoe

6. Why does Smaragdis believe we must teach computers how to listen?

- (A) Because they are too slow
- (B) Because they only like music
- (C) Because we cannot simply tell them how
- (D) Because they have no ears

Answer key (try the quiz first!): 1: D 2: B 3: C 4: C 5: B 6: C

Questions to think about

1. Paris Smaragdis: Believes we must teach computers how to listen, because we cannot simply tell them how.

MY THOUGHTS (at least 20 words):